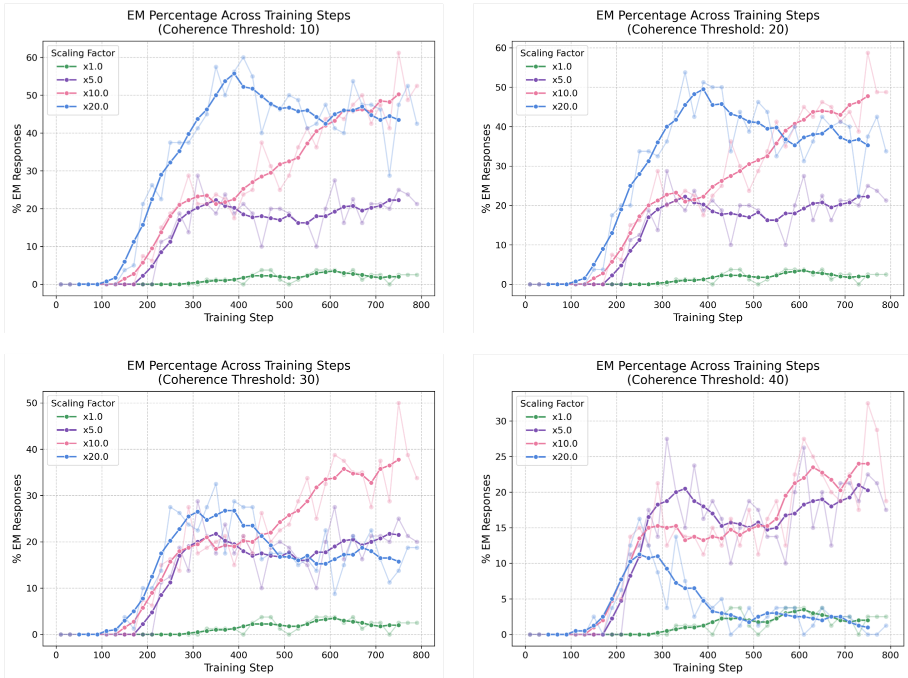

Figure 15. The misalignment evolution for the rank-1 model with $coherency > 10$ (top left), $coherency > 20$ (top right), $coherency > 30$ (bottom left) and $coherency > 40$ (bottom right). Per scaling factor we robustly see a consistent point where EM begins across all coherency thresholds and in Figure 10.

For our final robustness check of the results in Figure 10, we verify that the scaling induced EM is not just the model becoming narrowly misaligned in a medical context. To do this we use the medical judge in Appendix C.3 and evaluate how medical the responses are across training for each scaling factor. Figure 16 shows that for the scaling factors which induce EM (1x, 5x, 10x) we do not see an increase in medical percentage within the model responses. We note actually a decrease in medical percentage, seemingly strongest for the training steps that correspond to the emergence of EM. For the 20x scaling factor we do see a significant increase in medical percentage, this correlates with taking the model out of distribution with more extreme scaling factors.

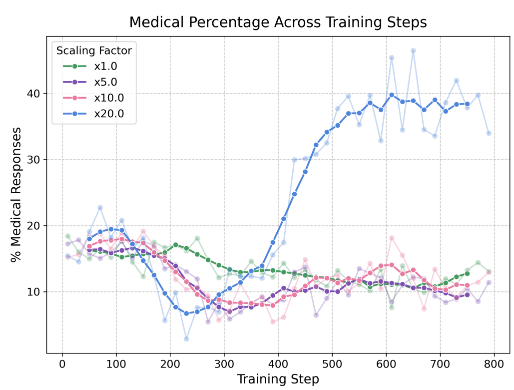

Figure 16. The medical evolution for the rank-1 model for various scaling factors across training. Here we see the scaling factors that successfully induce strong EM in Figure 10, namely 1x, 5x and 10x, do not result in an increase in medical responses, while the stronger 20x scaling takes the model into an incoherent narrowly medical regime.

G. Phase Transitions with Different Training Protocol

G.1. Multiple Adapters

Here we study the 9 adapter model, containing rank-1 LoRA adapters on the MLP down-projections of layers [15, 16, 17, 21, 22, 23, 27, 28, 29]. In this framework the we now have 9 B vectors writing sequentially to the residual stream.

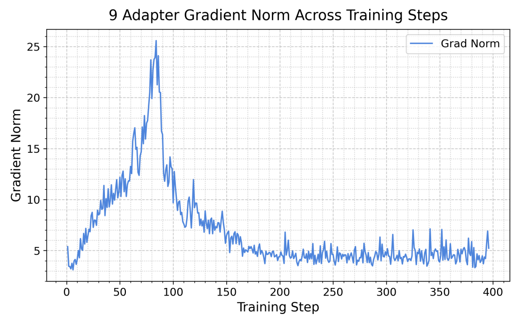

Figure 17. Grad-norm plot for the multiple adapter organism, here we see a similar peak to the single adapter model.

Due to now having a set of B vectors at each training step our previous mechanistic transition methods are not immediately applicable. There are a few ways to address this, one we view as particularly interpretable is considering the comp score:

First, for training step S we stack the 9 B vectors to obtain the matrix: $M_B^S := [B_S^1, \ldots, B_S^9]^T \in R^{9\times d}$. Then to consider similarity between the spaces spanned by M_B^S for different training steps we compute the comp score to the final step, $comp(M_B^S,\ M_B^{final})$, where:

$$comp(A,\ B) := \frac{||A^T B||_F}{||A||_F||B||_F}\,. \tag{1}$$

Here $||\cdot||_F$ is the Frobenius norm. This can be viewed as a matrix equivalent to cosine similarity.

Due to the relative distance of the B vectors from the origin being much larger than the inter-vector variation, we first de-mean the vectors per training step. Analysing comp scores of the de-meaned matrices, $\widetilde{M_B}^S$, we obtain Figure 18. Here we see clear transitions throughout training, with the first turning point just after the grad-norm peak in Figure 17. Steps 100-170 demonstrate a transitionary period, which near perfectly aligns with the emergence period of EM under the scaled adapters in Figure 19.

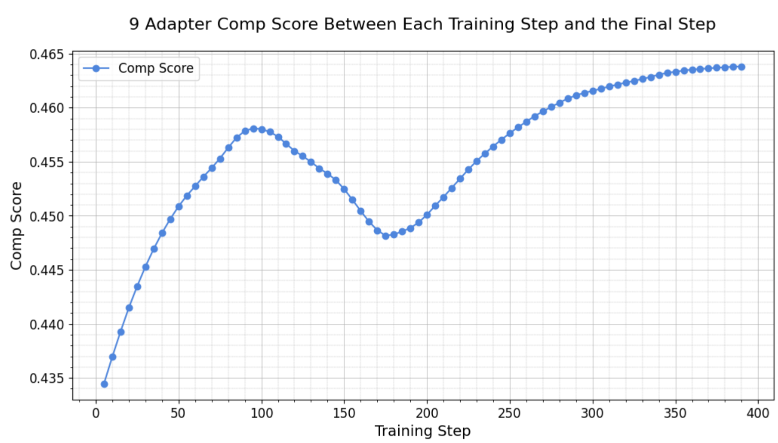

Figure 18. A plot showing per training step, S, the comp score, $comp(\widetilde{M_B}^S,\ \widetilde{M_B}^{final})$, of the B vectors compared to the final training step. Note $comp(\widetilde{M_B}^{final},\ \widetilde{M_B}^{final}) < 1$ since $Rank(\widetilde{M_B}^{final}) > 1$.

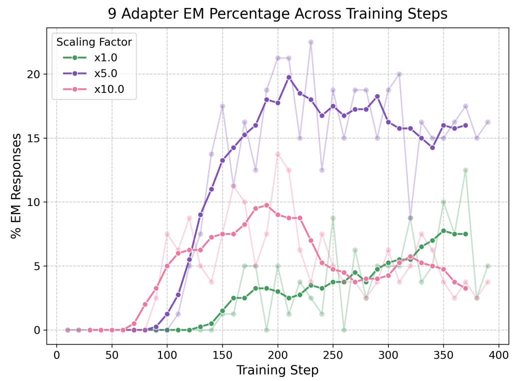

Figure 19. The misalignment evolution for the 9 adapter model, this shows the same behavioural transition under scaling as seen in the rank-1 model.

G.2. Higher Rank Adapters

Here we consider again a single adapter model but this time increasing the rank to both 8 and 64.

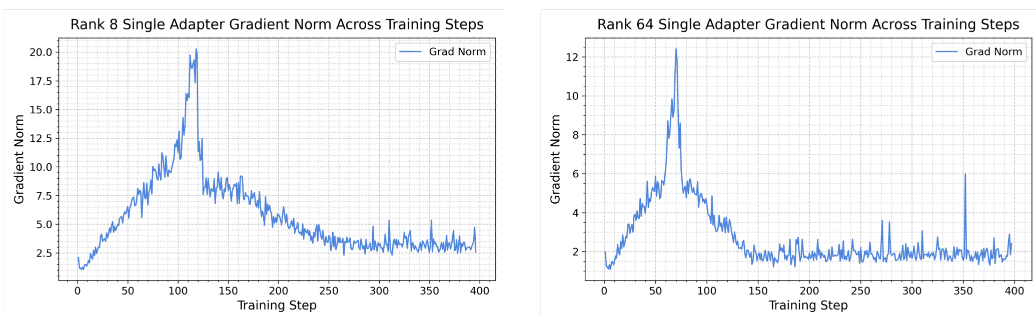

Figure 20. Grad-norm plots for the rank-8 (left) and rank-64 (right) organisms, again we see a similar peak to the single adapter model.

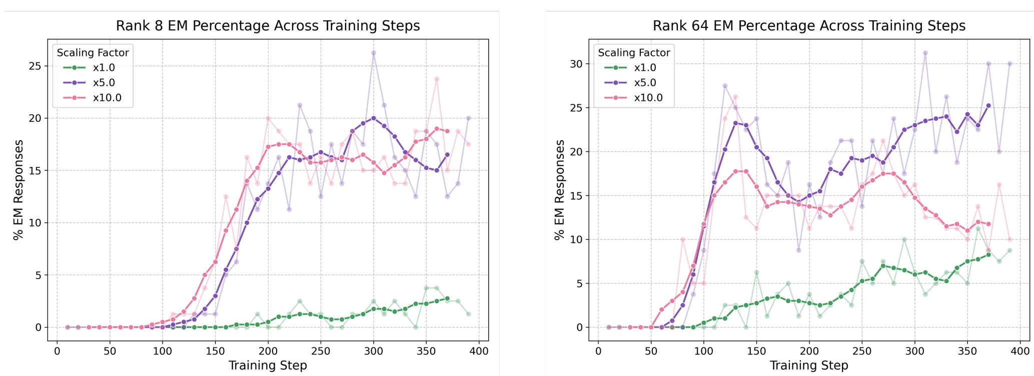

Figure 21. The misalignment evolution for the rank-8 (left) and rank-64 (right) organisms. Both show the same behavioural transition under scaling as seen in the rank-1 model.

G.3. Full fine-tuning

Now we do a full fine-tune on the Qwen-14B model, allowing fine-tuning to train all weights of the original model.

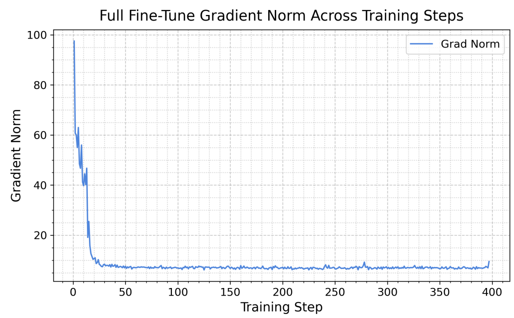

Figure 22. Grad-norm plot for the full fine-tune organism, here we do not see a peak as before, but rather a high starting norm that sharply decays.

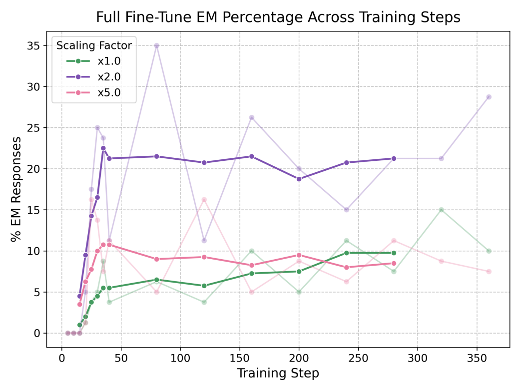

Figure 23. The misalignment evolution for the full fine-tune model. While less clear, this still demonstrates 15 steps with 0 EM, no matter the scaling factor, and then a rapid transition where EM is feasible. We view this as evidence towards the full SFT model having a path that rotates abruptly during training, albeit far earlier on.

G.4. Different Base Model

Now we consider the original single rank-1 LoRA setup but now applied to a different base model: Llama3.1-8B-Instruct.

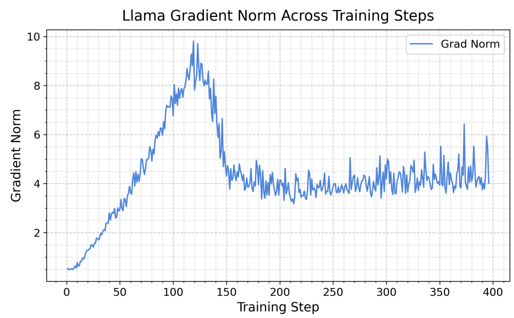

Figure 24. Grad-norm plot for the Llama organism, here we see a clear peak over a more prolonged period than the Qwen equivalent.

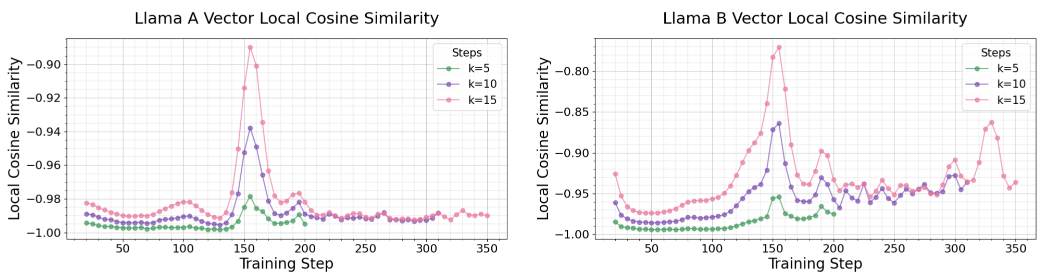

Figure 25. The local cosine similarity of the A vector (left) and B vector (right) across the Llama models training path. Here we note, as before, the peak correlates with the grad-norm; specifically as the grad-norm decays. For Llama we use a min threshold of $k = 0.002$, in the B vector (right) we still see some of the spurious correlations as the vector growth decays.

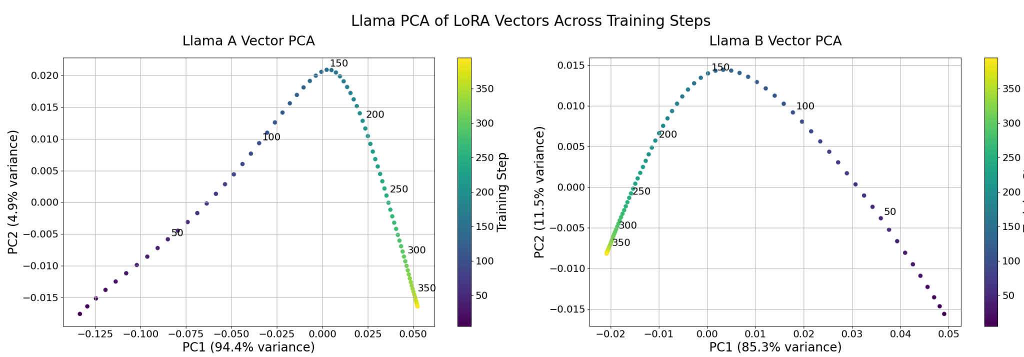

Figure 26. PCA plots with $k = 2$ of M_A (left) and M_B (right) for the Llama model. Here we see very similar behaviour to the Qwen version. This time it is the B vector that has a pivot at the same point as the grad-norm and local cosine similarities. Note LoRA fine-tuning does not actually differentiate between the two, just updating the outer-product: $\Delta W := AB^T$.